SQL Detective Game
Unity Seminar – Project Proposal

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Introduction

Our project is a 3D educational game designed to teach SQL query-building through story-driven gameplay. The game immerses players in a dynamic virtual environment where they take on the role of an investigator navigating a data-rich world. By solving SQL-based puzzles, players will learn how to construct queries, analyse tables, and retrieve meaningful data to progress in the game.

Purpose Of The Project

In today's world, where the volume of data grows exponentially alongside our reliance on it, the need to effectively manage, retrieve, and analyse information has become a critical challenge. This creates an essential goal: enabling the effective navigation, retrieval, and analysis of data to unlock its full potential, but accessing it requires foundational skills.

Many people interested in entering the world of data lack experience and need an introduction to SQL and data management. Furthermore, teaching children to navigate and retrieve information from large datasets from an early age is essential for their personal development and crucial for society.

Our project addresses these gaps by providing an immersive, story-driven educational experience that teaches SQL query-building through gameplay. By simulating a rich data environment, players will learn to navigate databases, construct queries, and extract insights in a hands-on manner.
In addition, It has already been proven that teaching a subject in the form of a fun activity such of a game has a greater impact on remembering the materials and understanding them, which is why this game has an edge over other non-interactive SQL teaching materials.

Target Audience

Our target audience includes a diverse group of people new to SQL who need a platform that immerses them in an authentic environment, allowing them to feel more connected to the data they work with, as well for children, highlighting the importance of introducing data literacy and SQL skills at an early age.
Additionally, the game appeals to casual gamers who enjoy story-driven thinking games, blending entertainment with educational value.

Market Analysis

The number of SQL-based educational games available on the market is limited,
but SQL Detective stands out with key differences:

1. 3D Environment: Unlike most SQL games, which are 2D or text-based, our game places players in an immersive 3D environment, creating a realistic and engaging experience.

2. Dynamic Clues and Story-Driven Scenarios: The game features evolving missions and clues that adapt to the player's progress, integrating SQL tasks naturally into the storyline. This ensures continuous engagement and learning.

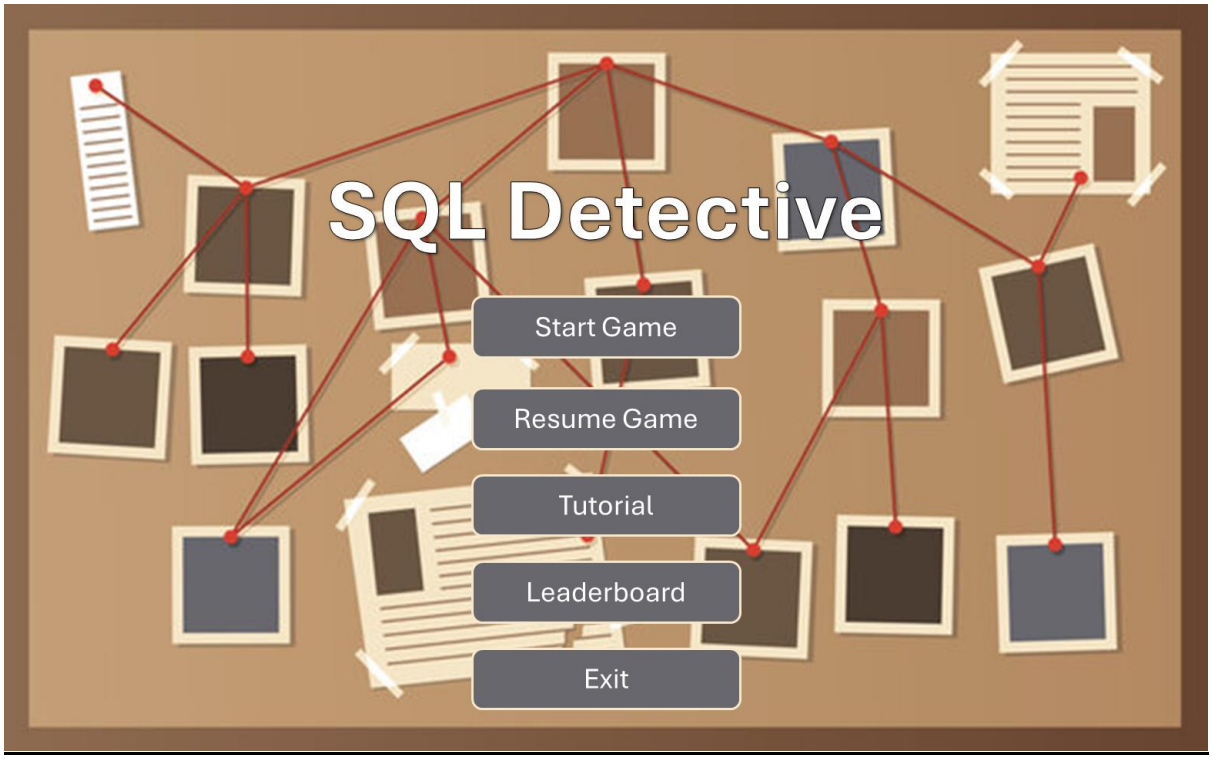
3. Cross-Platform Interaction: Players solve puzzles by interacting with the game world on the PC while using their phone for query-building. This unique approach enhances immersion and adds technological complexity.

4. Replayability and Modularity: The game's modular framework allows new datasets and scenarios to be added easily, enabling continuous learning and long-term engagement

Screen Structure

At the start, Main Menu will be represented and include:

1. Start Game
   1.1. Start
   1.2. Choose difficulty
2. Continue saved game
3. Tutorials
4. View Leaderboard
5. Exit



Gameplay

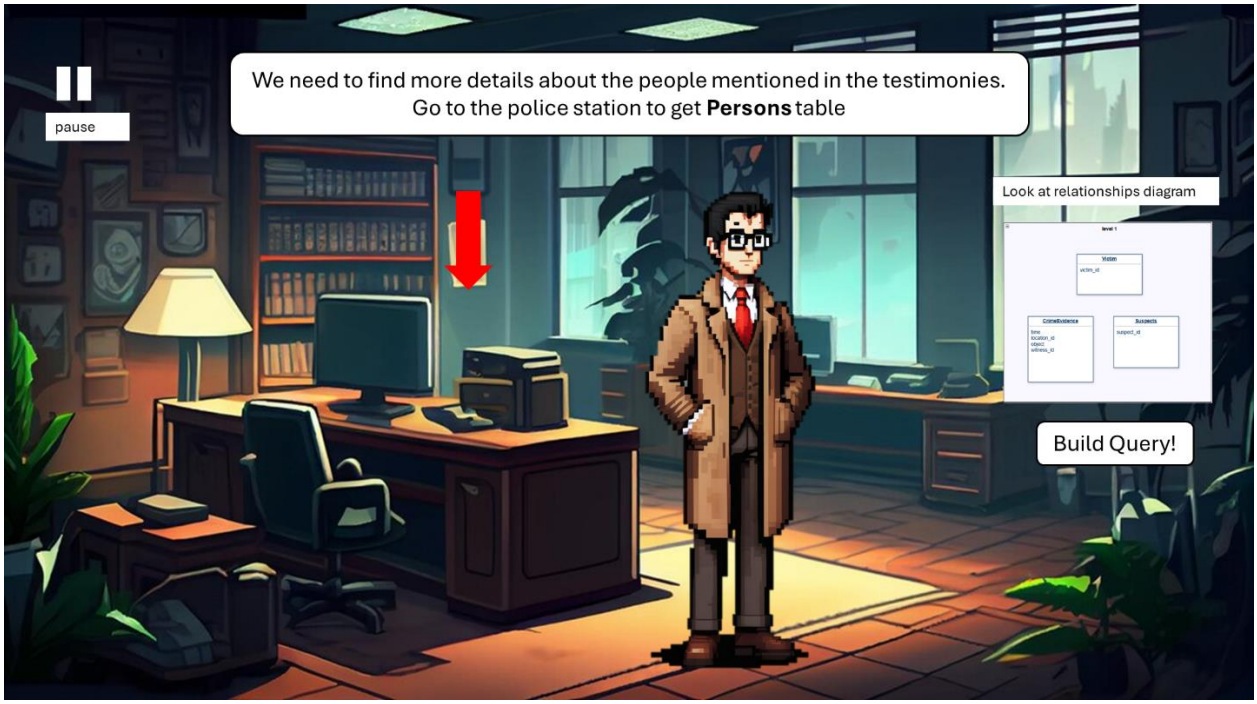
Players assume the role of a detective solving intricate crime mysteries. Each session revolves around investigating a unique case, such as solving a murder or uncovering criminal activities.
Players explore a richly detailed 3D environment, collecting clues and interacting with objects to uncover vital information. The gameplay is centered around solving SQL-based puzzles to interrogate dynamic databases and progress through the story.

The databases evolve dynamically as the player progresses. At the beginning, the player is introduced to one simple table, such as CrimeEvidence, accompanied by a narrative explanation.

The player starts with basic tasks like SELECT and FROM queries to retrieve simple data, and progresses to more advanced challenges, such as JOIN operations and filtering with WHERE clauses.

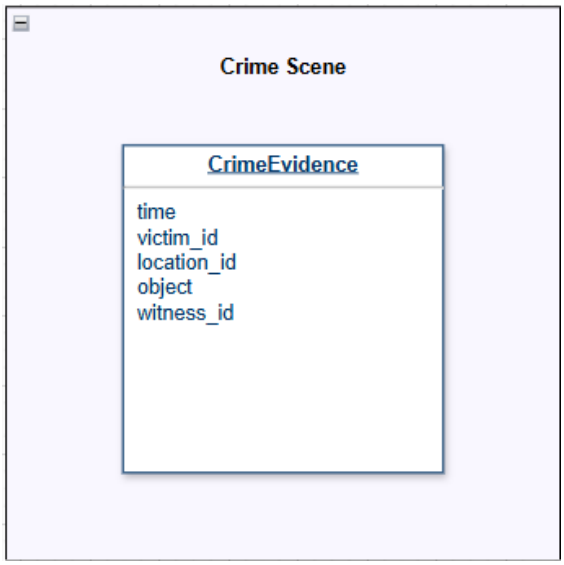
As the player unlocks new tables, the schema dynamically expands, requiring increasingly complex queries and critical thinking.

The detective is guided on what to do next, such as being directed to the police station to access the Persons table, with visual clues like an arrow pointing to the correct door. This progression ensures players develop confidence and advanced SQL skills while staying immersed in the narrative.

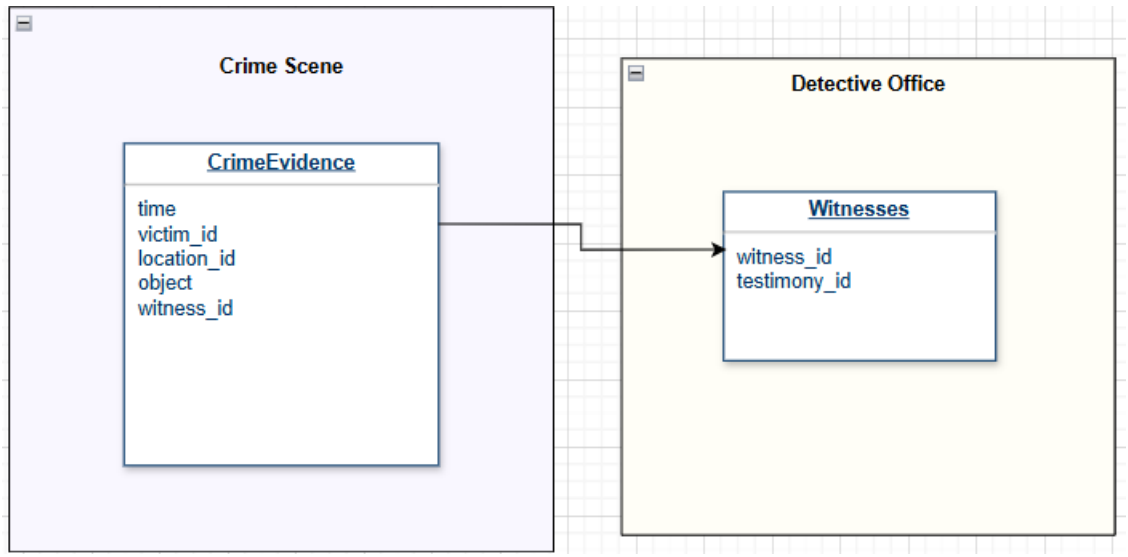


For example:

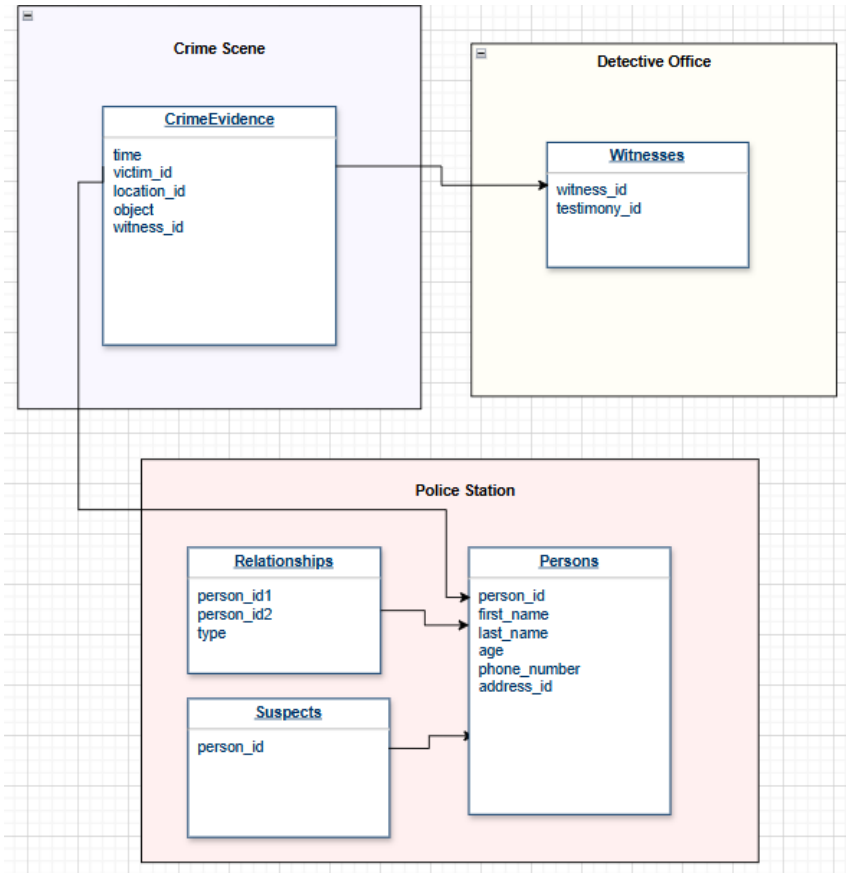
Level 1: The player starts with CrimeEvidence table and builds basic queries to retrieve data about the crime.
after correct answer, the player will move to detective's office.



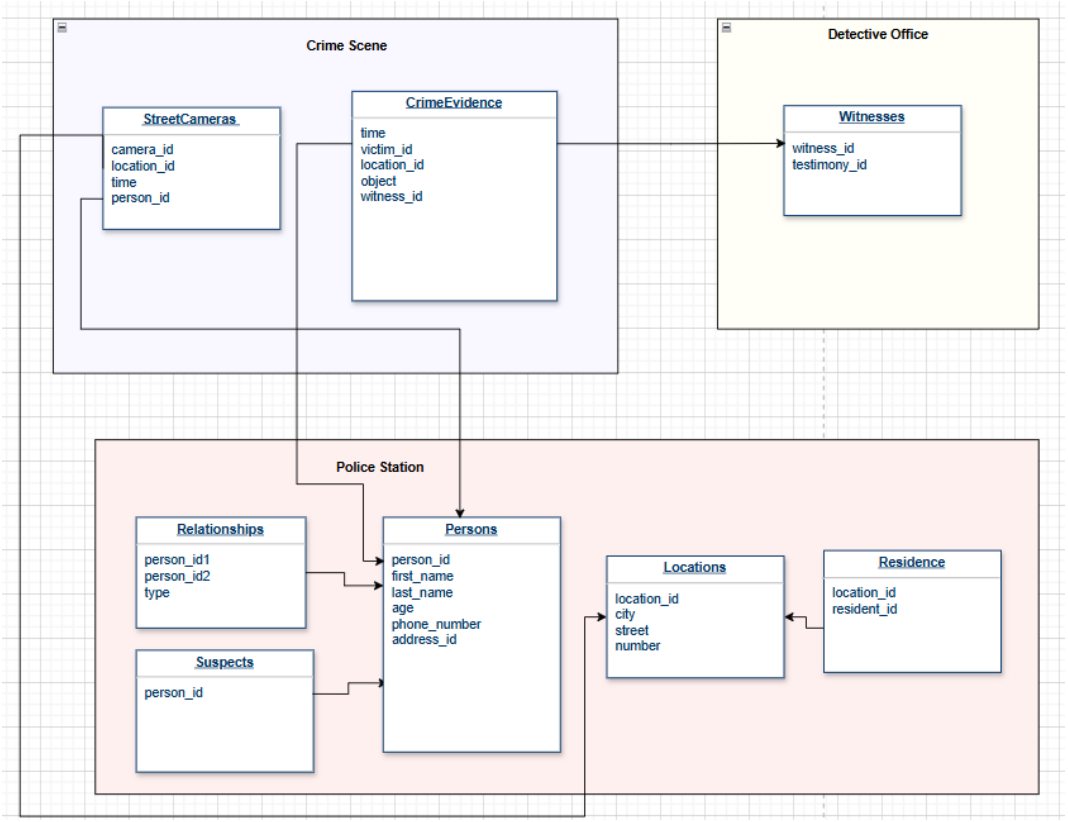
Level 2: After moving to the detective's office, the Witnesses table is introduced. The player learns to join tables to match evidence with witness testimonies.



Level 3: In the police station, additional tables such as Persons, Suspects and Relationships are unlocked. The player now faces multi-table queries to identify connections between suspects, witnesses, and victims, and filtering on wanted fields.



On next levels, more tables are unlocked such as Locations, Residence, and StreetCameras.



SQL mechanics

The game progresses through solving questions using SQL data retrieval, seamlessly blending education and gameplay. The SQL mechanics are designed to provide a hands-on, interactive learning experience while challenging the player's logical and problem-solving skills.

Platform Integration

The system operates on two interconnected platforms:
1. PC: Displays the main game environment, the evolving schema, and query results. Players interact with the narrative and explore the crime scene or detective's office for clues.
2. mobile: Serves as the SQL query-building platform, allowing players to interact directly with the database to construct queries and manage data.

Phone Interface Features

The phone acts as the core tool for database interaction. The interface includes the following components:

1. **Query Panel:** An empty panel for constructing SQL queries, with an "EXECUTE" button to run the retrieval operation.
2. **Reserved Words Panel:** Displays essential SQL keywords (SELECT, FROM, WHERE, JOIN, aggregate functions, etc.).
3. **Options List:** Shows available tables and fields for the player to query.
   Once a table is selected, its fields are displayed for players to drag into the query panel.

Query Construction

Players can construct their queries by dragging reserved words and tables to the query panel.
After selecting SELECT and FROM, players can choose fields from the selected table to display, and add conditions like WHERE clauses or aggregate functions.

The Suspects Table

In addition to query-building, players are provided with a "suspects" table, which serves as a dynamic tool for tracking and managing their investigation. Players can:
1. Add Records: Use INSERT INTO to add individuals they consider suspicious.
2. Delete Records: Use DELETE to remove individuals they rule out as suspects.
3. Retrieve Records: Query the suspects table to guess the final answer - who they believe committed the crime. Incorrect guesses result in losing a life and deducting points based on the scoring system, adding stakes to the player's choices.

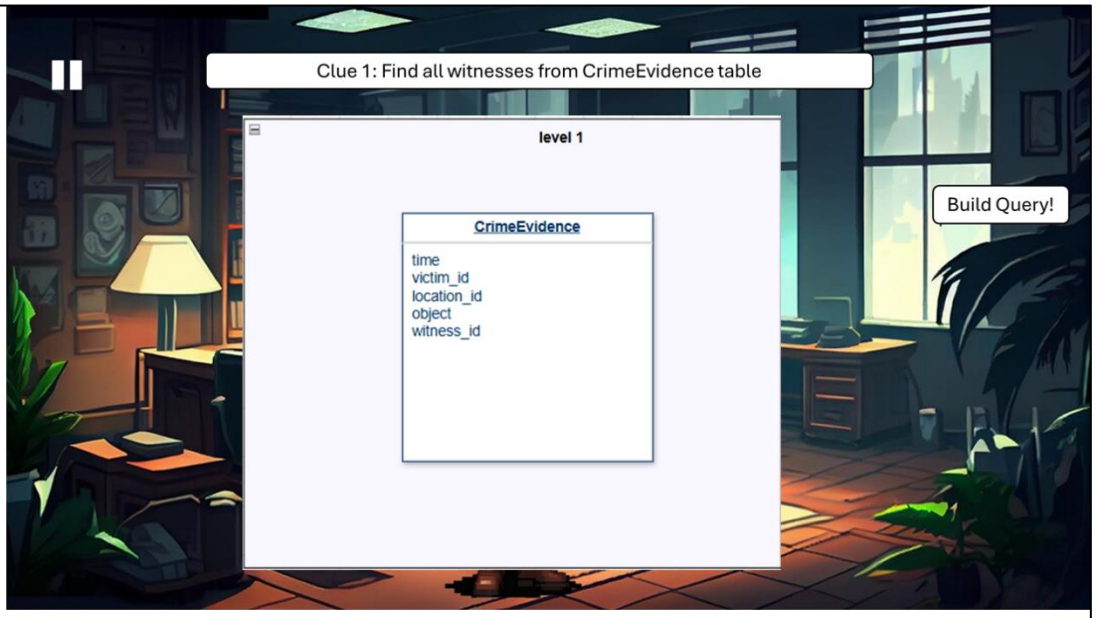
Schema Presentation on PC

The SQL mechanics use the PC for game display and schema visualization, while the phone handles query-building. Players start with simple tables like CrimeEvidence and expand the schema as new levels introduce advanced SQL concepts. Query results displayed on the PC drive the narrative, reinforcing learning through gradual progression and immediate feedback.

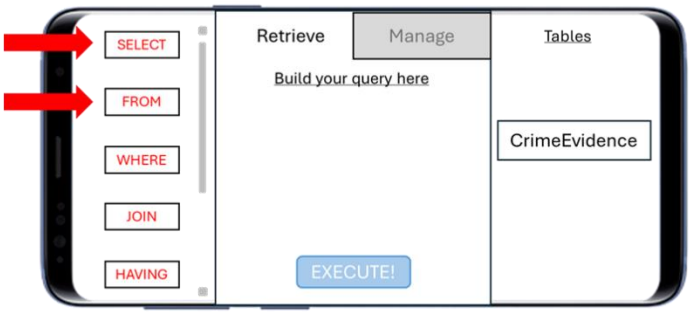
The player starts in his initial place, and has a task to do

as clicking on SQL query mode on mobile, the screen on PC show the current scheme and table.

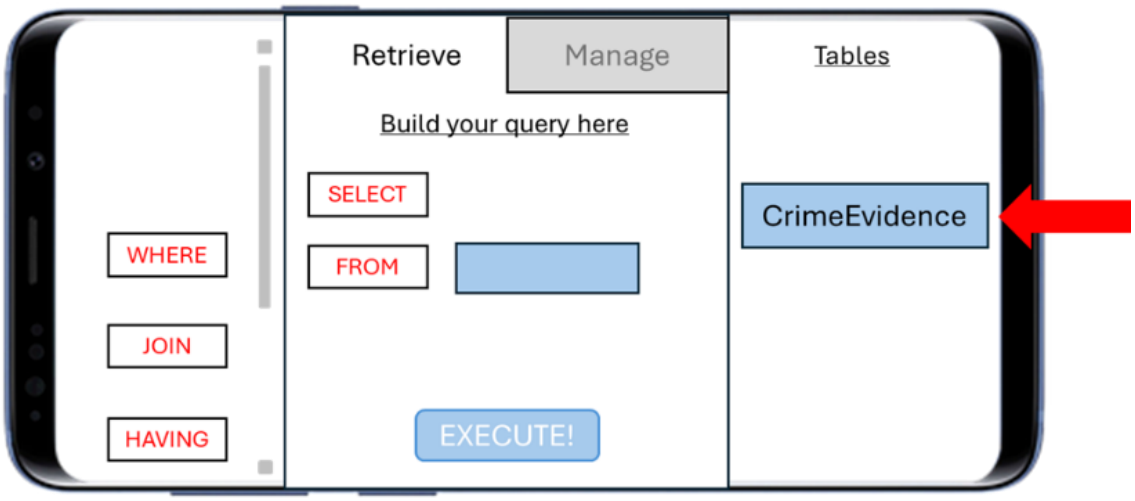
PC:



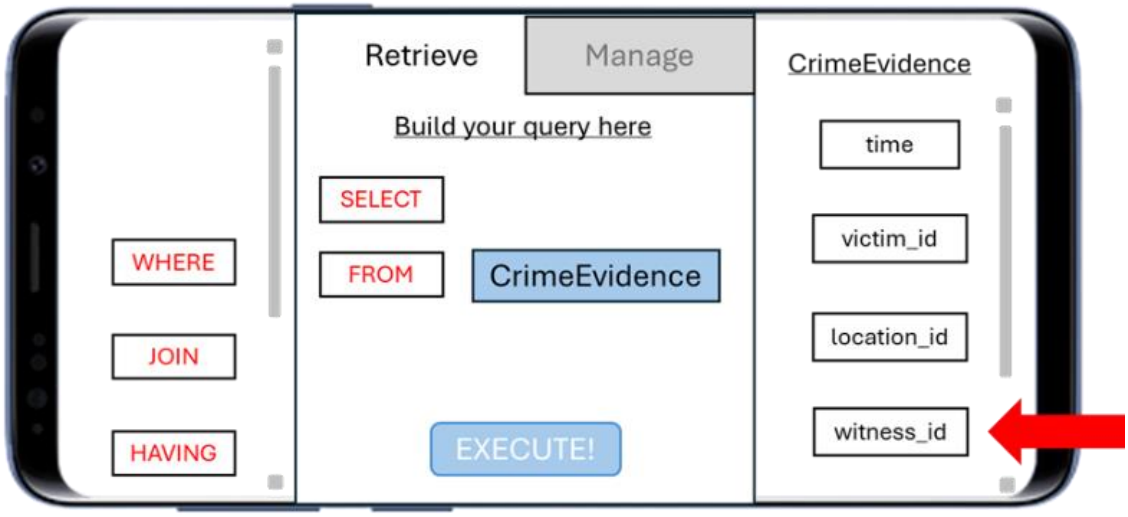
Phone:



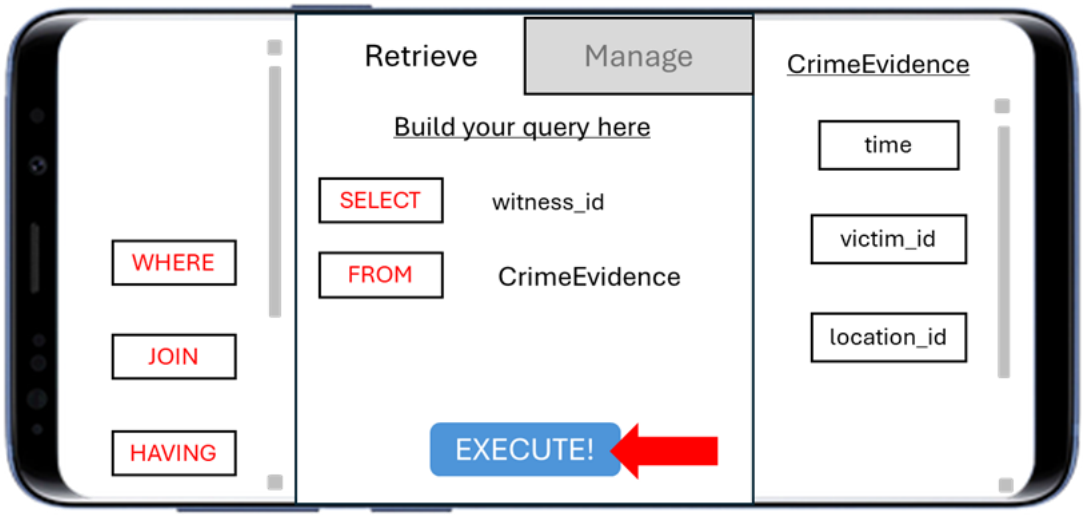
After dragging the From clause, a list of available tables is shown on right side



After choosing the table, its fields are shown on the right side.



After building a valid query, the player can click on Execute botton



The output is shown on PC and an answer (correct or incorrect).

Scoring and Lives

The scoring system incentivizes efficiency and accuracy. Scores are calculated based on:
1. Time: The quicker the query is solved, the higher the score.
2. Attempts: Fewer attempts bring better scores.
3. Query Complexity: Advanced queries earn bonus points.

Players have a limited number of lives, lost when making incorrect guesses about the final suspect. This system adds stakes to gameplay, encouraging thoughtful and accurate SQL execution.

Cloud

The game uses cloud integration to make sure progress is never lost. Player's progress, including completed levels, scores, and achievements, are saved in the cloud, so players can continue from where they left off on any device.
The cloud also stores dynamic scenarios and leaderboard data, encouraging competition and replayability.

Features

Dynamic Schema Evolution

As players progress, the schema dynamically expands to include new tables and relationships, teaching SQL concepts incrementally. For example, starting with CrimeEvidence and gradually introducing Witnesses, Relationships, and Locations tables allows players to tackle increasingly complex queries.

Interactive Query-Building Interface

The phone interface allows players to construct SQL queries through an intuitive drag-and-drop system. This dual approach supports both beginners and advanced users, while the PC displays the evolving schema and query results.

Cross-Platform Gameplay

The integration between PC and phone ensures seamless gameplay. Players explore the 3D environment and manage clues on the PC while using their phones to interact with the database and solve puzzles.

Replayability and Modular Design

The game's modular framework allows new puzzles to be added easily, ensuring replayability and continuous learning.

Technologies Used

Unity Game Engine

The project is built using Unity, a powerful engine for developing cross-platform 3D games. Unity enables us to create an immersive and visually detailed environment while supporting gameplay mechanics like query-building, schema progression, and interactive exploration. Its scripting capabilities in C# allow seamless integration with databases and real-time game logic.

SQL Database Integration

SQL Server is used to manage the game's data backend. This includes dynamic schema evolution, real-time query execution, and database updates as players progress. It ensures that query-building tasks are realistic and tied to actual database operations.

Cloud Integration

Cloud services are leveraged for saving player progress, achievements, and scores. This ensures continuity across devices and allows dynamic updates, such as adding new puzzles or datasets, without requiring local changes.

C# Programming

All core game logic is implemented in C#. This includes query validation, gameplay mechanics, scoring systems, and real-time interaction between the phone and PC. C# provides robust tools for building scalable and interactive game systems.

Mobile-Desktop Synchronization

Real-time communication protocols are used to synchronize the PC (which displays the main environment and schema) with the phone (which handles query-building). This ensures seamless cross-platform gameplay and interaction.